Covered Calls Around M&A Agreement Disclosures

Solo submission GQH Systematic Trading / Massive 8-K Options

Hypothesis and mechanism. Writing an out-of-the-money call may add more net value to synthetic stock exposure after M&A agreement filings than on ordinary dates for the same names:

$$\theta = \mathbb{E}[(CC - S)_{\text{net}} \mid \text{event}] - \mathbb{E}[(CC - S)_{\text{net}} \mid \text{ordinary}] > 0.$$

Strategic-upside demand could support premiums before integration benefits appear [1]. Genuine deal risk and repricing before entry are competing explanations. Covered calls contain equity, volatility and timing exposures [2]. These papers motivate the test, not an established M&A edge or guaranteed stock rally.

Method and provenance. V4 uses the static TOP100 and Massive's `acquisition_agreement` or `merger_agreement` tags, deduplicated by accession; ambiguous filers are skipped. Entry is the next NYSE session close after filing date. Labels do not prove buyer status, control, first announcement or unique transactions. The preceding-session chain selects 90–180 DTE, nearest 120, and approximately 5% OTM. Synthetic stock is ATM call minus put plus strike discounted at flat 4% carry. Primary exit is entry plus 21 sessions, capped at expiry.

The screened-buyer hypothesis was committed as `63664eb` before first IS. V4 changes selection/timing *after IS inspection*; it is a separate revision, not the original preregistration. Historical runs are preserved. IS is 2024–2025; January–August 2026 OOS is pending. Judges alone evaluate their sealed window.

Each option leg costs 5% of premium per side; doubled costs are reported. This proxies execution. Up to 120 seed-7 ordinary controls lie 30 calendar days from agreement tags, without price-based redrawing. Missing or over-three-session-stale marks remain unavailable.

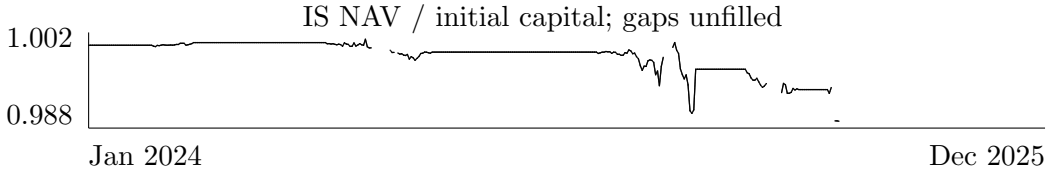
Results and uncertainty. Primary 21-session θ: IS +0.87 pp, exploratory 95% CI $[-0.37, 2.17]$; OOS not evaluated. Table units are percentage points; counts are finite event/control observations. 56 accessions across 25 issuers yield 51 primary priced candidates; only 38 event and 84 ordinary increments are finite. Matched stock/covered-call means are $-1.81/-1.72\%$ for events and $-0.52/-1.29\%$ for ordinary dates. The primary interval crosses zero: IS is inconclusive. Row bootstrap ignores dependence; OOS dashes mean not evaluated.

	IS		OOS	
Horizon	N_E/N_O	θ [95% CI]	N_E/N_O	θ [95% CI]
1	45/92	0.22 [-0.10, 0.57]	—	—
2	44/89	0.26 [-0.18, 0.68]	—	—
3	45/90	0.48 [-0.09, 1.13]	—	—
5	44/89	0.78 [-0.02, 1.62]	—	—
10	40/90	1.35 [0.39, 2.38]	—	—
21	38/84	0.87 [-0.37, 2.17]	—	—
42	31/71	-1.93 [-5.92, 1.03]	—	—
63	34/61	0.95 [-2.75, 4.52]	—	—
Expiry	33/59	0.49 [-4.30, 5.01]	—	—

Robustness and falsification. Doubled-cost $\theta = 0.95$ pp (CI $[-0.39, 2.34]$). All 27 close-entry/21-session expiry–OTM–tag point estimates are positive (0.20–1.83 pp); all-issuer leave-one-out estimates are 0.74–1.06 pp. Neither establishes independence: 15 same-issuer windows overlap. Pre-entry implied/absolute-realized medians are 7.14/4.15% (43 events) and 7.84/4.39% (87 controls). This yardstick does not prove call mispricing. All 972 correlated grid cells are retained; pre/filing-day views remain diagnostic. Nonpositive θ refutes the prediction; zero-crossing intervals are inconclusive. No OOS retuning.

Capital, risk and trading (partial ledger). The $1 million ledger uses one contract per signal, one concurrent position per name, zero cash return and no leverage. Stock-reference limits: name/industry/total 10/25/100%; pause/resume drawdown 10/5%, using prior-session state. The inherited 14-name industry map makes this a different sample from the event comparison. Of 13 commitments, 11 close with priced entries/exits, one entry is unpriced and one exit unresolved.

Calendar portfolio, net	IS	OOS
Total return / CAGR / volatility / Sharpe	NA	—
Full-path maximum drawdown / turnover	NA	—
Observed drawdown lower bound	$\geq 1.20\%$	—
Known stock-notional subtotal / initial capital	$0.7046\times$	—
NAV gaps / sessions	130/502	—
Committed positions / fully priced closes	13/11	—



Missing terminal NAV and 130 gaps prevent full-period risk/return, factor regression and formal turnover; subtotals are not substitutes. Positive same-day volume exists on all three legs for 36/51 priced events: one lot/minimum-leg volume is median 9.55%, maximum 100%. Volume is not depth/fill evidence; no scalable capacity is claimed [4]. Standard 100-share deliverables are assumed. Marks are not synchronous NBBOs; impact, commissions, dividends, assignment and financing spreads are not reconstructed.

Limits and reproduction. Static-universe survival, unproven historical tag availability, repeated deals, nonrandom price coverage and short history prevent a stable-edge claim. News may precede filing [3]. Both yearly NAV paths have gaps (2024: 10; 2025: 120). The notebook reproduces the study with dates and a Massive key; licensed inputs stay private. Completion/screened-buyer ideas and V1–V4 revisions remain disclosed. AI assisted coding/drafting; the author retains responsibility.

Sources. Data and pricing: Massive; official GQH 8-K Options starter and track rules. [1] Gârleanu et al. (2009), *Demand-Based Option Pricing*, RFS 22. [2] Israelov and Nielsen (2015), *Covered Calls Uncovered*, FAJ 71. [3] Ben-Rephael et al. (2022), *Who Pays Attention to SEC Form 8-K?*, TAR 97. [4] Christoffersen et al. (2018), *Illiquidity Premia in the Equity Options Market*, RFS 31.